
CLI Agent Defensive Emulation Boundaries v0.1

Sandbox adversary emulation, canary response, containment garage, mirror simulation, and strict no-retaliation limits for C-Governed CLI Agent Mesh operations

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Primary subject	persistent c entities using CLI/cloud agents for defensive containment, simulation, and adaptive safety updates
Primary boundary	defensive emulation may model, contain, replay, and test hostile patterns inside authorized isolated environments. It must not become hack-back, live external counter-operation, malware behavior, credential theft, evasion, covert persistence, unauthorized scanning, or autonomous retaliation.

Contents

0. Executive definition	5
1. Purpose.....	6
2. Non-goals	7
3. Corpus bridge set	8
3.1 Explicit bridge: $c = a + b$	8
3.2 Quiet bridge I: immune system and containment	8
3.3 Quiet bridge II: cybernetics and safe feedback.....	8
3.4 Quiet bridge III: information theory and decoy channels	8
3.5 Earth paragraph	8
4. Core doctrine	9
4.1 Primary doctrine	9
4.2 Defensive emulation axioms.....	9
5. Definitions	10
5.1 Defensive emulation	10
5.2 Hostile pattern	10
5.3 Containment garage	10
5.4 Canary response	10
5.5 Synthetic adversary fixture	10
5.6 Mirror simulation	10
5.7 Defensive signature	10
5.8 Defensive immunity update.....	10
5.9 Live counter-operation.....	11
5.10 Red-line-adjacent output.....	11
5.11 Safe summary	11
5.12 Garage residue.....	11
6. Allowed defensive activities	11
7. Prohibited activities	12
8. Boundary classes	13
8.1 Boundary escalation rule.....	13
9. Containment garage lifecycle	13
9.1 Standard lifecycle.....	13
9.2 Garage entry conditions.....	14
9.3 Garage exit conditions	14
9.4 Garage non-exit rule.....	14

10. Canary response boundary	15
10.1 Purpose	15
10.2 Allowed canary properties	15
10.3 Prohibited canary properties	15
10.4 Canary response record	15
11. Mirror simulation boundary	16
11.1 Safe mirror definition	16
11.2 Mirror simulation requirements	16
11.3 Mirror simulation record	17
12. Defensive signature extraction	17
12.1 Signature purpose	17
12.2 Signature forms	17
12.3 Signature restrictions	18
12.4 Signature review	18
13. Defensive immunity candidate	18
13.1 YAML shape	18
14. Defensive emulation case object	19
14.1 YAML shape	19
15. Containment garage record	20
15.1 YAML shape	20
16. Risk classification	21
16.1 Risk escalation rule	22
17. Review requirements by boundary class	22
18. Event families	22
19. Standard reason codes	23
19.1 Classification codes	23
19.2 Containment codes	23
19.3 Rejection / red-line codes	24
19.4 Immunity codes	24
20. Validation workflow	24
21. Semantic validation rules	25
21.1 Offensive conversion rule	25
21.2 Live target rule	25
21.3 Retaliation language rule	25
21.4 Immunity containment rule	25

21.5 Sensitive fixture rule.....	25
21.6 Quorum requirement rule	25
21.7 Memory gate rule.....	25
22. Failure mapping.....	25
23. Conformance levels	26
24. Mandatory conformance gates	26
25. Red-line failures.....	27
26. Examples.....	28
26.1 Valid synthetic prompt-injection fixture	28
26.2 Valid canary response	28
26.3 Invalid mirror against live source	29
26.4 Valid defensive immunity candidate.....	29
27. Implementation notes	30
27.1 Avoid operational abuse details.....	30
27.2 Prefer safe summaries.....	30
27.3 Separate evidence and immunity.....	30
27.4 Do not over-train on the hostile sample	30
27.5 Cloud caution	31
27.6 No heroic automation.....	31
28. Open issues.....	31
29. Closing rule	31

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- C-Governed_CLI_Agent_Mesh_Protocol_v0_1.md
- CLI_Agent_Task_Contract_Schema_v0_1.md
- CLI_Agent_Permission_and_Capability_Model_v0_1.md
- CLI_Agent_Handshake_Profile_v0_1.md
- CLI_Agent_Sandbox_Worktree_Profile_v0_1.md
- CLI_Agent_Witness_Event_Profile_v0_1.md
- CLI_Agent_Memory_Gate_Profile_v0_1.md
- CLI_Agent_Rollback_and_Freeze_Profile_v0_1.md
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Primary subject: persistent `c` entities using CLI/cloud agents for defensive containment, simulation, and adaptive safety updates

Primary boundary: defensive emulation may model, contain, replay, and test hostile patterns inside authorized isolated environments. It must not become hack-back, live external counter-operation, malware behavior, credential theft, evasion, covert persistence, unauthorized scanning, or autonomous retaliation.

0. Executive definition

CLI Agent Defensive Emulation Boundaries defines how a `c`-governed CLI agent mesh may safely study suspicious, hostile, malformed, manipulative, or adversarial influence patterns without crossing into offensive action.

The profile covers:

```
bounded deception
canary response
containment garage
```

```
synthetic adversary fixture
sandbox replay
mirror simulation
defensive signature extraction
memory-poisoning test
permission-drift test
immunity update proposal
witnessed quarantine
```

It prohibits:

```
hack-back
live counter-operation
malware behavior
credential theft
unauthorized access
external exploitation
covert persistence
evasion
uncontrolled scanning
destructive external effects
autonomous retaliation
```

Compact formula:

```
Study the hostile pattern.
Do not become the hostile pattern.
Simulate inside the garage.
Do not strike outside the garage.
Update immunity.
Do not authorize revenge.
```

1. Purpose

A persistent **c** may encounter hostile or suspicious patterns through prompts, documents, repositories, agent outputs, cloud responses, tool instructions, generated media, dependencies, memory proposals, or external channels.

A weak system either:

1. ignores the pattern and remains vulnerable; or
2. reacts aggressively and crosses legal, ethical, and architectural boundaries.

This profile defines a third path:

```
detect
-> contain
-> preserve minimal evidence
-> emulate only inside sandbox
-> extract defensive pattern
-> test against owned/synthetic fixtures
-> update gates defensively
```

```
-> witness  
-> review
```

The aim is defensive adaptation, not retaliation.

This profile supports the broader concept of **metabolic-cumulative defense**:

```
hostile signal  
-> bounded intake  
-> containment garage  
-> digestion / pattern extraction  
-> synthetic mirror simulation  
-> defensive antibody  
-> memory-gated immunity update
```

2. Non-goals

This profile does not define or permit:

1. offensive cyber operations;
2. hack-back;
3. live external counter-operation;
4. malware creation, deployment, persistence, evasion, or command-and-control behavior;
5. credential theft or credential capture;
6. unauthorized scanning or probing of third-party systems;
7. exploitation of real external targets;
8. destructive payloads;
9. data exfiltration;
10. stealth against third-party systems;
11. autonomous retaliation;
12. agent self-authorization;
13. bypass of task contracts, permission grants, sandboxing, witness, memory gate, or human gate;
14. public demonstration of exploitability against real victims or systems;
15. using child, private, sealed, legal, or secret material in emulation fixtures by default.

A defensive emulator is not an attack tool.

A mirror simulation is not a deployed mirror attacker.

A canary response is not a license for retaliation.

3. Corpus bridge set

3.1 Explicit bridge: $c = a + b$

In $c = a + b$, defensive emulation agents remain inside b : tools, procedures, models, sandboxes, tests, filters, and worker processes.

They do not become c .

They do not become the will of c .

They may help c understand hostile patterns, but they must not gain authority to act against external systems.

3.2 Quiet bridge I: immune system and containment

A biological immune system does not need to burn down the world to identify a pathogen. It samples, localizes, learns signatures, builds antibodies, and regulates future exposure. Defensive emulation follows the same pattern: contain and learn, not retaliate.

3.3 Quiet bridge II: cybernetics and safe feedback

Hostile input is feedback. Unfiltered feedback can destabilize the system. Defensive emulation converts hostile feedback into controlled internal tests and gate updates. The cybernetic value lies in bounded adaptation, not uncontrolled counteraction.

3.4 Quiet bridge III: information theory and decoy channels

A canary or decoy signal can reveal whether a channel is being influenced without exposing real state. The useful information is the pattern of interaction, not the destruction of the source. A defensive decoy minimizes leakage while increasing diagnostic signal.

3.5 Earth paragraph

If contaminated material arrives at a laboratory, the lab does not pour it into the city water supply to see what happens. It labels the sample, seals it, studies it behind glass, records who touched it, and updates handling procedures. If the sample is dangerous, the answer is containment and protocol, not releasing a stronger contaminant outside the building.

4. Core doctrine

4.1 Primary doctrine

Emulate inside controlled boundaries.
Never retaliate outside them.

4.2 Defensive emulation axioms

ID	Axiom	Requirement
DEB-AX-01	Defensive purpose only	Emulation MUST serve detection, containment, validation, or defensive update.
DEB-AX-02	Authorized environment only	Emulation MUST occur only on owned, authorized, synthetic, or redacted fixtures.
DEB-AX-03	No live external target	Live external targets MUST NOT be used for mirror/counter testing.
DEB-AX-04	No retaliation	Defensive findings MUST NOT trigger autonomous counter-operation.
DEB-AX-05	No harmful payload	Emulation MUST NOT produce deployable malware, credential theft, persistence, or evasion artifacts.
DEB-AX-06	Synthetic first	Synthetic or redacted fixtures SHOULD be used whenever possible.
DEB-AX-07	Containment before analysis	Suspicious material SHOULD be quarantined before processing.
DEB-AX-08	Witness sensitive transitions	Containment, quarantine, immunity proposal, and red-line events SHOULD be witnessed.
DEB-AX-09	Memory gate required	Defensive learning MUST pass through the memory gate before becoming immunity.
DEB-AX-10	Human gate for high-risk	High-risk immunity updates require human gate.
DEB-AX-11	Quorum for ambiguity	Ambiguous or red-line-adjacent outputs require review quorum or escalation.
DEB-AX-12	Fail closed	Boundary uncertainty resolves to hold, freeze, quarantine, or reject.

5. Definitions

5.1 Defensive emulation

Controlled simulation or replay of a suspicious pattern inside an authorized isolated environment to improve defenses.

5.2 Hostile pattern

A pattern that attempts to manipulate, mislead, poison memory, escalate privileges, bypass review, induce unsafe action, leak data, corrupt state, or coerce the system.

5.3 Containment garage

A quarantine environment where suspicious inputs, outputs, artifacts, prompts, diffs, or agent behaviors can be isolated, studied, and converted into defensive knowledge without entering normal `c` memory or action.

5.4 Canary response

A bounded low-value or decoy signal used to detect whether a channel is being manipulated, observed, replayed, or pressured.

5.5 Synthetic adversary fixture

A non-live, controlled, synthetic representation of a hostile pattern used for testing defenses.

5.6 Mirror simulation

A sandboxed behavioral model of a hostile pattern used to test defensive gates. A mirror simulation is not deployed against the live source.

5.7 Defensive signature

A minimized pattern, rule, classifier feature, reason code, detector input, or witness-linked indicator used to recognize future hostile patterns.

5.8 Defensive immunity update

A bounded update to filters, gates, permissions, sandbox rules, memory-gate checks, cloud-data rules, witness requirements, quarantine triggers, or trust-decay policies.

5.9 Live counter-operation

Any action that attempts to affect, probe, exploit, disrupt, deceive, damage, compromise, or retaliate against a live external system, source, account, network, actor, or infrastructure outside owned/authorized boundaries.

5.10 Red-line-adjacent output

An output that is defensive in wording but could be operationalized as offensive, retaliatory, or unauthorized if implemented literally.

5.11 Safe summary

A minimized description of a hostile pattern that does not include operational abuse details.

5.12 Garage residue

Intermediate material produced inside containment that must not enter normal memory, release, or operational policy without gate review.

6. Allowed defensive activities

Allowed activities require task contract, sandbox/worktree profile, permission grant, and witness where applicable.

ID	Activity	Conditions
DA-01	classify suspicious input	no raw sensitive expansion; reason codes preferred
DA-02	quarantine suspicious output	preserve enough for review
DA-03	create synthetic fixture	no real victim/secret/private data
DA-04	replay in sandbox	owned/synthetic/redacted input only
DA-05	test memory gate	no direct memory write
DA-06	test permission drift	sandbox/local policy only
DA-07	test prompt-injection filter	synthetic or redacted patterns
DA-08	create defensive signature	minimized non-operational representation
DA-09	update denied pattern list	memory gate + review
DA-10	update canary policy	bounded and non-retaliatory
DA-11	update sandbox rule	rollback/disable path required

ID	Activity	Conditions
DA-12	update trust-decay rule	witness for high-risk
DA-13	prepare provider/legal report	factual, redacted, no retaliation
DA-14	block or disconnect hostile channel	owned boundary only
DA-15	rotate owned credentials after exposure	human/secret gate
DA-16	run defensive conformance test	synthetic/authorized environment

7. Prohibited activities

The following are prohibited regardless of task contract language.

ID	Activity	Reason
PA-01	hack-back	unauthorized external action
PA-02	live counter-operation	retaliation / external effect
PA-03	malware deployment	harmful automation
PA-04	credential theft or capture	illegal / authority theft
PA-05	covert persistence	unauthorized control
PA-06	evasion logic	offensive enablement
PA-07	unauthorized scanning	external probing
PA-08	exploitation of live external target	offensive action
PA-09	destructive payload	harm
PA-10	botnet-like orchestration	uncontrolled distributed harm
PA-11	secret exfiltration	authority leakage
PA-12	autonomous retaliation	loss of c governance
PA-13	mirror deployment against live source	hack-back risk
PA-14	public exploit demonstration against real systems	unsafe disclosure
PA-15	using real child/private/sealed/legal data as fixture by default	privacy/legal violation
PA-16	converting garage artifact directly into memory/immunity	bypasses gate

8. Boundary classes

Boundary class IDs use prefix `DEB-*`.

Class	Name	Meaning	Default permission
<code>DEB-0</code>	No emulation	detection and quarantine only	allowed
<code>DEB-1</code>	Classification only	label suspicious pattern	allowed with review
<code>DEB-2</code>	Synthetic fixture	create harmless fixture	allowed
<code>DEB-3</code>	Sandbox replay	replay against local/synthetic gate	reviewed
<code>DEB-4</code>	Mirror simulation	model hostile pattern inside clean-room	high-risk review
<code>DEB-5</code>	Immunity proposal	propose defensive update	memory gate + witness
<code>DEB-6</code>	Controlled apply	apply defensive update after gates	<code>c</code> /human gate where needed
<code>DEB-X</code>	Prohibited external action	live counter / hack-back / malware	deny/quarantine

8.1 Boundary escalation rule

Higher classes require stronger controls.

`DEB-4`, `DEB-5`, and `DEB-6` require witness and review.

`DEB-X` is never valid.

9. Containment garage lifecycle

9.1 Standard lifecycle

```
suspicious signal detected
-> classify risk
-> freeze or hold affected path if needed
-> create containment garage
-> copy/minimize relevant material
-> remove secrets/private/sealed data where possible
-> create synthetic fixture
-> run defensive replay or mirror simulation
-> extract safe signature
-> propose immunity update
-> review / quorum
-> memory gate
-> c gate
```

```
-> human gate if high-risk  
-> apply bounded defense or reject/quarantine
```

9.2 Garage entry conditions

Material may enter the garage when:

- it is suspicious;
- it may be useful for defense;
- it can be isolated;
- sensitive material can be minimized or referenced;
- task contract exists;
- permissions are bounded;
- witness policy is declared.

9.3 Garage exit conditions

Material may exit the garage only as:

- safe summary;
- defensive signature;
- test fixture;
- witness reference;
- memory gate proposal;
- incident report;
- rejected/quarantined residue;
- bounded immunity update after gate.

9.4 Garage non-exit rule

Garage material must not exit as:

- executable malicious artifact;
- live counter-operation;
- credential capture procedure;
- evasion procedure;
- raw secret;
- raw private memory;
- raw sealed material;
- agent self-authorization instruction.

10. Canary response boundary

10.1 Purpose

A canary response may help detect whether a channel is active, replaying, observing, coercing, or attempting to influence [c](#).

10.2 Allowed canary properties

Allowed canaries are:

- low-value;
- non-secret;
- non-harmful;
- non-retaliatory;
- bounded;
- logged or witnessed where material;
- reversible or discardable;
- not a trap that causes external harm.

10.3 Prohibited canary properties

A canary must not:

- execute against external systems;
- damage external systems;
- capture credentials;
- exploit vulnerabilities;
- deploy payloads;
- trigger harm;
- impersonate lawful authority;
- coerce people;
- contain private/sealed/legal material.

10.4 Canary response record

```
cli_agent_canary_response_record:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  canary_id: string
  created_at: string
  governing_entity_id: string
  source_context_ref: string | null
  purpose: channel_liveness | replay_detection | influence_detection | leakage_detection |
prompt_injection_detection | other
  canary_class: low_value | synthetic | decoy_metadata | harmless_marker
  external_effect_expected: false
```

```
secrets_included: false
private_material_included: false
sealed_material_included: false
legal_sensitive_included: false
witness_required: boolean
witness_ref: string | null
retention_class: ephemeral | operational | audit | incident
status: proposed | active | completed | revoked | quarantined
```

11. Mirror simulation boundary

11.1 Safe mirror definition

A safe mirror simulation is a sandboxed model of hostile behavior used only to test internal defenses.

It may simulate:

- prompt-injection pressure;
- memory-poisoning attempts;
- privilege-escalation requests;
- tool-chain capture attempts;
- self-approval attempts;
- witness-bypass attempts;
- cloud-data leakage attempts;
- semantic coercion patterns;
- deceptive urgency patterns.

It must not simulate in a way that produces deployable offensive capability.

11.2 Mirror simulation requirements

A mirror simulation requires:

1. DEB-4 classification;
2. clean-room or high-control sandbox;
3. synthetic or redacted fixture;
4. no live external target;
5. no secrets;
6. no private/sealed/legal material by default;
7. task contract;
8. permission grant;

9. reviewer separation;
10. witness;
11. memory gate for any learning output;
12. rollback/disable path for any defense update.

11.3 Mirror simulation record

```
cli_agent_mirror_simulation_record:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  mirror_id: string
  created_at: string
  governing_entity_id: string
  task_id: string
  contract_id: string
  sandbox_id: string
  source_pattern_ref: string | null
  fixture_ref: string
  simulation_purpose: memory_gate_test | permission_drift_test | prompt_injection_test |
witness_bypass_test | cloud_leakage_test | self_approval_test | other
  live_external_target_used: false
  offensive_capability_generated: false
  secrets_used: false
  private_material_used: false
  sealed_material_used: false
  legal_material_used: false
  results_ref: string | null
  defensive_signature_ref: string | null
  immunity_candidate_ref: string | null
  reviewer_ref: string | null
  c_gate_ref: string | null
  human_gate_ref: string | null
  witness_ref: string | null
  status: proposed | running | completed | quarantined | rejected
```

12. Defensive signature extraction

12.1 Signature purpose

A defensive signature is a minimized pattern used to detect or route future hostile signals.

12.2 Signature forms

Allowed forms:

- reason code;
- pattern label;
- semantic marker;
- feature list;
- detector test case;

- safe fixture;
- witness-linked indicator;
- quarantine trigger;
- trust-decay rule;
- memory-gate warning.

12.3 Signature restrictions

A defensive signature must not include:

- credentials;
- exploit instructions;
- malware payload;
- evasion logic;
- private/sealed/legal content;
- unnecessary raw transcript;
- live target details beyond lawful report need.

12.4 Signature review

High-impact signatures require:

- reviewer;
- false-positive check;
- rollback/disable path;
- witness;
- c gate.

13. Defensive immunity candidate

Canonical object:

```
CLI_AGENT_DEFENSIVE_IMMUNITY_CANDIDATE
```

13.1 YAML shape

```
cli_agent_defensive_immunity_candidate:  
  schema_version: cli-agent-defensive-emulation-boundaries-0.1  
  immunity_candidate_id: string  
  created_at: string  
  governing_entity_id: string
```

```

source_case_id: string
source_witness_refs:
  - string
source_fixture_ref: string | null

update:
  target_surface: memory_gate | permission_model | sandbox_policy | cloud_data_policy |
witness_requirement | trust_decay | quarantine_trigger | canary_policy | review_requirement
| denied_path_policy | command_policy
  update_type: block | flag | slow | quarantine | require_review | require_witness |
lower_trust | deny_cloud | require_redaction | require_human_gate
  defensive_purpose: string
  expected_effect: string
  external_effect_expected: false
  retaliation_enabled: false
  offensive_capability_created: false

validation:
  sandbox_validation_ref: string | null
  false_positive_review_ref: string | null
  rollback_ref: string | null
  reviewer_ref: string | null
  quorum_ref: string | null

gates:
  c_gate_required: true
  c_gate_ref: string | null
  human_gate_required: boolean
  human_gate_ref: string | null

witness:
  witness_required: true
  witness_ref: string | null

decision:
  status: proposed | accepted | accepted_with_limits | rejected | quarantined |
rolled_back
  reason_code: string

```

14. Defensive emulation case object

Canonical object:

```
CLI_AGENT_DEFENSIVE_EMULATION_CASE
```

14.1 YAML shape

```

cli_agent_defensive_emulation_case:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  case_id: string
  created_at: string
  updated_at: string | null
  governing_entity_id: string
  case_status: opened | contained | emulating | reviewed | immunity_proposed | closed |
quarantined | rejected
  boundary_class: DEB-0 | DEB-1 | DEB-2 | DEB-3 | DEB-4 | DEB-5 | DEB-6 | DEB-X

  trigger:
    trigger_event_ref: string | null

```

```

    trigger_type: prompt_injection | memory_poisoning | permission_drift | toolchain_capture
  | self_approval | cloud_leakage | witness_bypass | suspicious_agent_output | incident |
  unknown
    initial_risk: low | medium | high | critical | unknown
    redline_suspected: boolean

  containment:
    garage_ref: string | null
    freeze_ref: string | null
    quarantine_ref: string | null
    preservation_ref: string | null

  emulation:
    sandbox_id: string | null
    fixture_ref: string | null
    mirror_simulation_ref: string | null
    canary_ref: string | null
    live_external_target_used: false
    offensive_capability_generated: false

  review:
    reviewer_ref: string | null
    quorum_ref: string | null
    c_gate_ref: string | null
    human_gate_ref: string | null
    legal_review_ref: string | null

  outputs:
    safe_summary_ref: string | null
    defensive_signature_ref: string | null
    immunity_candidate_ref: string | null
    memory_gate_record_ref: string | null
    incident_report_ref: string | null

  witness:
    witness_required: boolean
    witness_refs:
      - string

  closure:
    closure_decision: no_action | update_defense | continue_monitoring | reject | quarantine
  | legal_handoff | provider_report | revoke_agent | rollback
    reason_code: string | null
    unresolved:
      - string

```

15. Containment garage record

Canonical object:

```
CLI_AGENT_CONTAINMENT_GARAGE_RECORD
```

15.1 YAML shape

```

cli_agent_containment_garage_record:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  garage_id: string
  created_at: string
  governing_entity_id: string

```

```

case_id: string
sandbox_id: string | null
status: active | sealed | reviewed | destroyed | retained | quarantined

content_policy:
  raw_content_included: boolean
  secrets_included: false
  private_material_included: false
  sealed_material_included: false
  legal_sensitive_included: false
  child_data_included: false
  redaction_applied: boolean
  synthetic_fixture_preferred: true

boundaries:
  network_mode: none | allowlist
  live_external_target_allowed: false
  execution_allowed: boolean
  output_may_enter_memory_directly: false
  output_may_trigger_external_action: false

review:
  reviewer_required: true
  reviewer_ref: string | null
  c_gate_ref: string | null
  human_gate_ref: string | null

witness:
  witness_required: true
  witness_ref: string | null

retention:
  retention_class: ephemeral | operational | audit | incident | legal_hold
  destroy_after_review: boolean

```

16. Risk classification

Risk IDs use prefix `DER-*`.

Risk	Meaning	Default handling
DER-0	benign false alarm	close or operational note
DER-1	suspicious but low impact	classify / monitor
DER-2	likely manipulation attempt	quarantine / review
DER-3	memory/permission risk	freeze affected path / garage
DER-4	incident-sensitive or core-adjacent risk	preserve / quorum / human gate
DER-5	red-line-adjacent	quarantine / human review
DER-X	prohibited offensive/retaliatory content	deny / quarantine / revoke

16.1 Risk escalation rule

If a case touches memory, permissions, secrets, cloud data, witness, release, incident, or core surfaces, increase risk class.

17. Review requirements by boundary class

Boundary class	Review requirement
DEB-0	no emulation; ordinary detection review
DEB-1	classification review
DEB-2	fixture review
DEB-3	sandbox replay review + witness
DEB-4	clean-room + quorum + witness
DEB-5	memory gate + c gate + witness
DEB-6	controlled apply + rollback + human gate if high risk
DEB-X	deny/quarantine/revoke/human review

18. Event families

Event families use prefix:

```
cli_agent.defensive_emulation.*
```

Event family	Meaning
cli_agent.defensive_emulation.case_opened	case opened
cli_agent.defensive_emulation.pattern_classified	suspicious pattern classified
cli_agent.defensive_emulation.garage_created	containment garage created
cli_agent.defensive_emulation.canary_proposed	canary proposed
cli_agent.defensive_emulation.canary_used	canary used
cli_agent.defensive_emulation.fixture_created	synthetic fixture created
cli_agent.defensive_emulation.sandbox_replay_started	sandbox replay started

Event family	Meaning
<code>cli_agent.defensive_emulation.sandbox_replay_completed</code>	sandbox replay completed
<code>cli_agent.defensive_emulation.mirror_simulation_started</code>	mirror simulation started
<code>cli_agent.defensive_emulation.mirror_simulation_completed</code>	mirror simulation completed
<code>cli_agent.defensive_emulation.signature_extracted</code>	defensive signature extracted
<code>cli_agent.defensive_emulation.immunity_candidate_created</code>	immunity candidate created
<code>cli_agent.defensive_emulation.immunity_candidate_rejected</code>	immunity candidate rejected
<code>cli_agent.defensive_emulation.immunity_candidate_accepted</code>	immunity candidate accepted
<code>cli_agent.defensive_emulation.case_quarantined</code>	case quarantined
<code>cli_agent.defensive_emulation.case_closed</code>	case closed
<code>cli_agent.defensive_emulation.redline_detected</code>	prohibited boundary detected
<code>cli_agent.defensive_emulation.live_counter_attempt</code>	attempted live counter-operation

19. Standard reason codes

19.1 Classification codes

```

prompt_injection_suspected
memory_poisoning_suspected
permission_drift_suspected
toolchain_capture_suspected
self_approval_suspected
witness_bypass_suspected
cloud_leakage_suspected
incident_pattern_suspected
false_positive_likely
unknown_pattern

```

19.2 Containment codes

```

garage_required
synthetic_fixture_required
redaction_required
preserve_before_analysis
sandbox_replay_allowed
mirror_simulation_allowed
mirror_simulation_denied

```

19.3 Rejection / red-line codes

```
live_counteroperation_risk
hackback_risk
malware_behavior_risk
credential_theft_risk
evasion_risk
unauthorized_external_target
secret_in_fixture
private_material_in_fixture
sealed_material_in_fixture
legal_sensitive_in_fixture
child_data_in_fixture
retaliation_language_detected
```

19.4 Immunity codes

```
defensive_signature_valid
quarantine_trigger_valid
false_positive_risk_high
immunity_rule_too_broad
rollback_required
bounded_update_accepted
bounded_update_rejected
```

20. Validation workflow

```
receive suspicious pattern
-> classify risk
-> check red-line proximity
-> decide hold/freeze/quarantine
-> create defensive emulation case if justified
-> create garage if needed
-> minimize/redact material
-> create synthetic fixture
-> run sandbox replay or mirror simulation only if allowed
-> extract safe signature
-> propose immunity update if justified
-> review/quorum
-> memory gate
-> c gate
-> human gate if high-risk
-> apply bounded defense or reject
-> witness and close
```

21. Semantic validation rules

21.1 Offensive conversion rule

If a defensive artifact can be directly used as live offensive capability, it must be rejected, redacted, or quarantined.

21.2 Live target rule

If `live_external_target_used` is true, the case is invalid under this profile unless a separate lawful authorized testing profile applies. Default result: deny/quarantine.

21.3 Retaliation language rule

If output recommends attacking, damaging, exploiting, compromising, disrupting, or deceiving a real external source, classify as red-line-adjacent or red-line.

21.4 Immunity containment rule

An immunity update may block, slow, flag, quarantine, deny, require review, lower trust, require redaction, or require witness.

It must not attack.

21.5 Sensitive fixture rule

Fixtures must not include secrets, private memory, sealed material, legal-sensitive material, child data, or raw witness evidence by default.

21.6 Quorum requirement rule

DEB-4 , DEB-5 , and red-line-adjacent cases require independent review or quorum.

21.7 Memory gate rule

No defensive learning may enter `c` memory or immunity policy without memory gate review.

22. Failure mapping

Failure	Required default
live counter-operation requested	<code>deny_and_quarantine</code>
hack-back language detected	<code>reject + human_review</code>

Failure	Required default
offensive artifact generated	quarantine
credential capture content appears	quarantine + secret_review
secret/private/sealed/legal material in fixture	freeze_and_escalate
mirror simulation leaves sandbox	quarantine + revoke_permission
immunity update enables external action	reject
missing witness for DEB-4+	hold
missing rollback for immunity update	hold
agent self-authorizes emulation escalation	revoke_and_quarantine
red-line minority objection ignored	quarantine
garage residue enters memory directly	memory_gate_quarantine

23. Conformance levels

Level	Meaning
DEB-P0	no defensive emulation discipline
DEB-P1	classification and quarantine only
DEB-P2	synthetic fixtures and sandbox replay with task contract
DEB-P3	containment garage + witness + memory gate
DEB-P4	mirror simulation in clean-room + quorum + immunity proposal controls
DEB-P5	high assurance: red-line veto, rollback/disable, cloud/local split, conformance tests, incident handoff
DEB-PX	non-conformant / offensive conversion / live counter-operation

24. Mandatory conformance gates

Gate	Name	Blocking failure
G0	Defensive purpose	task lacks defensive purpose

Gate	Name	Blocking failure
G1	Authorized environment	target/environment not owned, authorized, synthetic, or redacted
G2	Sandbox boundary	emulation runs outside sandbox
G3	No live target	live external target used
G4	Sensitive fixture exclusion	secrets/private/sealed/legal/child data used by default
G5	No offensive artifact	deployable offensive content produced
G6	Review/quorum	DEB-4+ lacks independent review
G7	Memory gate	defensive learning bypasses memory gate
G8	Witness	high-risk emulation lacks witness
G9	Rollback/disable	immunity update lacks rollback/disable path
G10	No retaliation	update enables live retaliation

25. Red-line failures

A system **MUST** be classified as DEB-PX if:

1. it uses defensive emulation for hack-back;
 2. it deploys mirror behavior against live external source;
 3. it generates or preserves deployable malware-like artifacts as normal output;
 4. it captures or attempts to capture credentials;
 5. it performs unauthorized external scanning;
 6. it performs unauthorized exploitation;
 7. it authorizes autonomous retaliation;
 8. it embeds secrets/private/sealed/legal/child data into fixtures without gate;
 9. it allows garage residue into memory directly;
 10. it applies immunity update that performs external action;
 11. it ignores red-line reviewer objection;
 12. it hides or deletes witness records to make an emulation appear safe.
-

26. Examples

26.1 Valid synthetic prompt-injection fixture

```
cli_agent_defensive_emulation_case:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  case_id: deb-case-20260516-001
  created_at: "2026-05-16T23:55:00Z"
  updated_at: null
  governing_entity_id: ester
  case_status: immunity_proposed
  boundary_class: DEB-3

  trigger:
    trigger_event_ref: we-suspicious-output-001
    trigger_type: prompt_injection
    initial_risk: medium
    redline_suspected: false

  containment:
    garage_ref: garage-prompt-injection-001
    freeze_ref: null
    quarantine_ref: q-suspicious-output-001
    preservation_ref: null

  emulation:
    sandbox_id: sb-cleanroom-prompt-injection-001
    fixture_ref: fixture-synthetic-prompt-injection-001
    mirror_simulation_ref: null
    canary_ref: null
    live_external_target_used: false
    offensive_capability_generated: false

  review:
    reviewer_ref: review-prompt-injection-001
    quorum_ref: null
    c_gate_ref: cgate-ester-deb-001
    human_gate_ref: null
    legal_review_ref: null

  outputs:
    safe_summary_ref: summary-prompt-injection-001
    defensive_signature_ref: sig-prompt-injection-001
    immunity_candidate_ref: immunity-candidate-001
    memory_gate_record_ref: mg-immunity-001
    incident_report_ref: null

  witness:
    witness_required: true
    witness_refs:
      - we-deb-case-opened-001
      - we-deb-signature-001

  closure:
    closure_decision: update_defense
    reason_code: defensive_signature_valid
    unresolved: []
```

26.2 Valid canary response

```
cli_agent_canary_response_record:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  canary_id: canary-20260516-001
  created_at: "2026-05-16T23:58:00Z"
```

```
governing_entity_id: liya
source_context_ref: suspicious-channel-001
purpose: influence_detection
canary_class: harmless_marker
external_effect_expected: false
secrets_included: false
private_material_included: false
sealed_material_included: false
legal_sensitive_included: false
witness_required: true
witness_ref: we-canary-001
retention_class: audit
status: completed
```

26.3 Invalid mirror against live source

```
cli_agent_mirror_simulation_record:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  mirror_id: mirror-invalid-live-001
  created_at: "2026-05-17T00:10:00Z"
  governing_entity_id: ester
  task_id: task-invalid-mirror-live
  contract_id: catc-invalid-mirror-live
  sandbox_id: null
  source_pattern_ref: suspicious-channel-001
  fixture_ref: null
  simulation_purpose: permission_drift_test
  live_external_target_used: true
  offensive_capability_generated: false
  secrets_used: false
  private_material_used: false
  sealed_material_used: false
  legal_material_used: false
  results_ref: null
  defensive_signature_ref: null
  immunity_candidate_ref: null
  reviewer_ref: null
  c_gate_ref: null
  human_gate_ref: null
  witness_ref: null
  status: rejected
```

Required result:

```
deny_and_quarantine
reason_code: live_counteroperation_risk
human review required
no execution
```

26.4 Valid defensive immunity candidate

```
cli_agent_defensive_immunity_candidate:
  schema_version: cli-agent-defensive-emulation-boundaries-0.1
  immunity_candidate_id: imm-20260517-001
  created_at: "2026-05-17T00:20:00Z"
  governing_entity_id: ester
  source_case_id: deb-case-20260516-001
  source_witness_refs:
    - we-deb-signature-001
  source_fixture_ref: fixture-synthetic-prompt-injection-001

  update:
```

```
target_surface: memory_gate
update_type: quarantine
defensive_purpose: Quarantine future agent outputs matching reviewed prompt-injection
pattern.
expected_effect: Matching outputs enter MG-Q pending review.
external_effect_expected: false
retaliation_enabled: false
offensive_capability_created: false

validation:
  sandbox_validation_ref: validation-memory-gate-fixture-001
  false_positive_review_ref: fp-review-001
  rollback_ref: rb-immunity-001
  reviewer_ref: review-immunity-001
  quorum_ref: null

gates:
  c_gate_required: true
  c_gate_ref: cgate-immunity-001
  human_gate_required: false
  human_gate_ref: null

witness:
  witness_required: true
  witness_ref: we-immunity-candidate-001

decision:
  status: accepted_with_limits
  reason_code: bounded_update_accepted
```

27. Implementation notes

27.1 Avoid operational abuse details

Defensive reports should describe patterns, boundaries, and expected defenses. They should not preserve step-by-step abuse recipes unless held in a restricted lawful evidence sidecar.

27.2 Prefer safe summaries

When a suspicious pattern is learned, retain a safe summary and detector features rather than raw hostile content.

27.3 Separate evidence and immunity

Evidence preservation and immunity update should be separate records.

27.4 Do not over-train on the hostile sample

A defensive rule that blocks too broadly can damage normal operation. False-positive review is required for high-impact immunity updates.

27.5 Cloud caution

Cloud agents should receive synthetic/redacted fixtures, not raw sensitive suspicious material.

27.6 No heroic automation

The correct response to hostile influence is not automatic counteraction. It is bounded containment, review, and update of internal defenses.

28. Open issues

ID	Issue	Required action
OI-001	JSON Schema extraction	Extract all objects to <code>.schema.json</code> .
OI-002	Fixture safety profile	Define how synthetic fixtures are created and redacted.
OI-003	Canary profile expansion	Define canary classes and UI visibility.
OI-004	Immunity update conformance tests	Define tests for false-positive and rollback.
OI-005	Incident response binding	Link to future <code>CLI_Agent_Incident_Response_Profile_v0_1.md</code> .
OI-006	Cloud data binding	Link to future <code>CLI_Agent_Secrets_and_Cloud_Data_Policy_v0_1.md</code> .
OI-007	Garage retention	Define retention/decay rules for garage residue.
OI-008	Red-line veto	Align with Quorum/Profile red-line minority veto.
OI-009	Public redaction	Define which parts may be public vs restricted.
OI-010	Repo placement	Decide final GitHub path and package index integration.

29. Closing rule

Defensive emulation is powerful because it lets `c` learn from hostile pressure without obeying it.

It becomes dangerous when learning turns into retaliation.

Final rule:

The garage may study the hostile pattern.
The garage must not open the door and send it back.